Muhammad Talha Khalid

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Research interests: liquid cooling of data centres; physics-informed digital twins; CFD of flow-induced vibration.

EDUCATION

Bachelor of Engineering, Mechanical Engineering Oct 2020 – Jun 2024
National University of Sciences and Technology (NUST), Islamabad

CGPA 3.70/4.00, top 5% of the cohort. Dean's Honours List in each of the last three semesters, for a top-ten CGPA.
Final year project: *Multi Axis Control Based Self Stabilizing Platform for Heavy Payloads*

RESEARCH EXPERIENCE

Research Assistant, Digital Twins for Data-Centre Cooling Jun 2026 – Present
NUST, Islamabad · supervised by Prof Ibraheem Haneef *Part-time*

- Writing a critical review of how close physics-informed digital twins are to real-time use in direct-to-chip and immersion-cooled data centres.
- Built a 184-paper corpus in 12 topic groups and compared the scope of 11 earlier reviews across 12 dimensions to locate the gap.
- Each published twin is graded against two scales set out in the review: a DT0 to DT6 maturity ladder, and a V0 to V5 scale for validation against measured data.
- Published twins, including ExaDigiT and DCVerse, end the cooling loop at a fixed cold-plate resistance with single-phase coolant. The review argues that boiling and conjugate heat transfer are the main missing physics.

Research Assistant, Computational Fluid Dynamics Mar 2026 – Present
NUST, Islamabad · supervised by Dr Emad Ud Din *Part-time*

- **Tandem flexible flags.** Ran a 36-case parametric study (a 6 × 6 grid of body-to-flag and flag-to-flag gaps, 0.25 to 1.5) at Re = 300 on the group's in-house immersed-boundary FSI solver (Fortran). It builds on the water-tunnel results reported by Mujtaba et al. (2021).
- Wrote Python scripts to process about 440 GB of output. They trim the start-up transient, check each run for solver problems and batch Tecplot figures of drag, lift, energy and flag motion.
- **Microfluidic cell sorter.** Building a COMSOL 6.4 model, based on a published design, of a channel that separates white from red blood cells by laminar flow and particle tracing. The 112-run study compares four channel designs, four wall-roughness factors and seven Reynolds numbers.

Remote Research Assistant, Laser Surface Texturing of Bearings Apr 2025 – Jun 2026
Qilu Institute of Technology, China · with Dr Jawad Aslam, on his postdoctoral project at Shandong University *Part-time*

- Prepared and formatted the research proposal and yearly plan, which set out circular, elliptical and triangular textures at 5–20% area density for friction tests on a tribology rig.
- Read and summarised earlier studies on textured 100Cr6 bearing steel, mostly for tapered roller bearing inner rings.

MANUSCRIPTS

1. Farzad, M. S., Haneef, I., **Khalid, M. T.**, Waris, R., Amin, F. and Khan, M. A. A. *Integrated Mechatronic Design of a Heavy-Payload Two-Axis Gimbal Under Cyclic Loading.* Manuscript in final revision.
2. **Khalid, M. T.** and Haneef, I. *Towards Physics-Informed Digital Twins for Data-Centre Liquid and Immersion Cooling Systems: A Review.* Manuscript in preparation.

ENGINEERING PROJECTS

Multi Axis Control Based Self Stabilizing Platform for Heavy Payloads Jun 2023 – May 2024
Final year project, team of four · project report · control system photos

- The team designed and built a two-axis (pitch and yaw) gimbal for heavy payloads. The frame was modelled in SOLIDWORKS, cast in aluminium and checked in ANSYS Mechanical for stress, deformation and fatigue.
- Responsible for the control subsystem. Developed the Arduino software for the 1,000 W BLDC servos, integrated MPU6050 feedback with a teammate's Kalman filter, tuned the PID loops and added fail-safes for the 40 A drivers.
- The follow-up manuscript (manuscript 1) tested the prototype with a 20 kg payload under a 200 N, 20 Hz cyclic load. It reports measured settling times of 0.38 s in yaw and 0.47 s in pitch, with overshoot of 1.2% or less.

Cooling-load estimate for a two-storey house Feb – Jun 2023
HVAC course project, team of five · Carrier HAP · project report

- Estimated the cooling load of each conditioned room in a two-storey house in Lahore, first in Carrier HAP and then by hand. In five of the six rooms compared, the hand figure came out 9–12% above HAP. The kitchen, where the stove dominates, went the other way: 19% below (22,362 against 27,600 Btu/hr).

Lunar electric rover: structural design without FEA Sep 2022 – Jan 2023
Mechanics of Materials course project, team of five · SOLIDWORKS, MATLAB · final design report · CAD files

- Designed a 12-wheel rover and checked its structure by hand, as FEA was not allowed. The calculations, scripted in MATLAB, covered an indeterminate chassis frame, lunar thermal stresses, a 1 m drop, column buckling and shaft fatigue. They showed 6061 aluminium falling short, so the design moved to 7068 with a Ti-6Al-4V suspension.

Smaller projects 2021

- An interactive Python stress-strain model, where temperature and cooling rate shift the modulus and yield strength through simple assumed factors (source code). Also an RC car with a self-designed motor driver (photos).

PROFESSIONAL EXPERIENCE

Product Operations Engineer, OneScreen Solutions May 2025 – Present
Islamabad · interactive flat panels for customers in the US, Canada, Europe and the Middle East *Full-time*

- Support, training, customer success and operations, mostly on US Eastern hours; supervise one junior team member.
- Wrote a Python panel-registration tool for the manufacturer's API; it saved 125+ hours on a 1,500+ panel rollout.
- Also wrote the scripts that now handle licence-key allocation, invoice capture from a desktop ERP with no API, and renewal and overdue-invoice reminders (Python, Google Apps Script, n8n).
- Documented 12+ internal workflows, which cut onboarding time for new staff by about 30%. Supported school-district rollouts in Duncanville and Lewisville, Texas, part of deployments worth over US$5 million.

Junior Wireline Engineer, Well Logging Energy (Pvt.) Ltd. Oct 2024 – Jan 2025
Field sites in Khyber Pakhtunkhwa *Full-time*

- Ran production logging (PLT) jobs at the Bettani, Togh and Nashpa wells, and set up a Power Automate flow for the tool inventory that cut retrieval time by about 20%.

Maintenance Engineering Intern, Attock Refinery Ltd. Jun – Jul 2024
Rawalpindi *Full-time*

- Four weeks across the refinery's maintenance sections and power plant. Assisted on pumps, compressors and heat exchangers, and took part in root cause analyses that used fault tree analysis and FMEA.

Engineering Intern, Sui Northern Gas Pipelines Ltd. (SNGPL) Aug – Sep 2022
Islamabad *Full-time*

- Most of the internship was spent with Corrosion Control on pipeline inspection, corrosion prevention and leak detection. Also visited compression stations and observed how SCADA monitors more than 9,000 km of pipeline.

Mathematics Solver, Photomath Dec 2021 – Jun 2022
Remote *Freelance*

- Solved and reviewed 100+ student problems, writing step-by-step solutions in LaTeX with GeoGebra figures.

LEADERSHIP AND SERVICE

Mentor Technical, ASHRAE NUST Student Branch Sep 2023 – May 2024
Treasurer of the same branch, Sep 2022 – Sep 2023

- Ran the branch's technical workshops, mentored junior members and secured US$5,400 in funding for their training. As treasurer, kept the budgets and records with the university finance office.

Volunteer, Islamabad schools and Khalq Foundation Jan – Jun 2023

- Raised funds for two drinking-water coolers at under-resourced schools near D-11, Islamabad, and taught the students Excel. Also helped run a clothing drive for flood-hit families (record).

HONOURS

Prime Minister's Merit Laptop Scheme, Government of Pakistan (2023). Third place, AKDC Design Challenge, NUST.

SKILLS AND LANGUAGES

Simulation: COMSOL Multiphysics, ANSYS Mechanical, MATLAB, Tecplot, Carrier HAP, immersed-boundary CFD
Design and hardware: SOLIDWORKS, AutoCAD, Arduino, LabVIEW, PID control
Programming and automation: Python, C++, Google Apps Script, REST APIs, n8n, Zapier, Power Automate, LaTeX
Languages: English (IELTS Academic 8.0, August 2026); Urdu (native)

REFEREES

Prof Ibraheem Haneef
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Dr Emad Ud Din
Former Head of Department, Mechanical Engineering, SMME, NUST · emaduddin@smme.nust.edu.pk

Dr Jawad Aslam
Formerly a professor at SMME, NUST; my supervisor at Qilu Institute of Technology, China · jawadaslam@smme.nust.edu.pk